CITS5508 Practice Exam — Chapter 7

Ensemble Learning and Random Forests

- This practice paper has **5 questions** for **60 marks**.
- Allow approximately **two minutes per mark** (≈ 2 hours total).
- Only UWA-approved calculators permitted.
- Answers are in the separate file *Practice Exam - Chapter 7 (Answer Key).md*.

Question 1. (12 marks)

(a) Explain, using the idea of the **law of large numbers** (or the "wisdom of the crowd"), why an ensemble of many **weak learners** can form a **strong learner**. *(4 marks)*

(b) This benefit relies on a critical assumption about the individual predictors. State it, and explain one practical way to make an ensemble's predictors satisfy it better. *(4 marks)*

(c) Explain the difference between a **hard voting** and a **soft voting** classifier, and state which tends to perform better and why. *(4 marks)*

Question 2. (12 marks)

(a) Explain the difference between **bagging** and **pasting**. *(3 marks)*

(b) Both methods generally give an ensemble with **similar bias but lower variance** than a single predictor. Explain why aggregation reduces variance. *(3 marks)*

(c) What is **out-of-bag (OOB)** evaluation? Approximately what fraction of the training set is OOB for a given predictor under bagging, and why is OOB evaluation useful? *(4 marks)*

(d) Why do bagging and pasting **scale well** across multiple CPU cores or servers, whereas boosting does not? *(2 marks)*

Question 3. (12 marks)

(a) Describe what a **random forest** is and the **extra source of randomness** it introduces compared with a plain bagging ensemble of decision trees. *(4 marks)*

(b) Describe **extra-trees (extremely randomized trees)**. What additional randomness do they add, and what are the consequences for **bias/variance** and **training speed**? *(4 marks)*

(c) Explain how a random forest computes **feature importance**, and give one practical use of it. *(4 marks)*

Question 4. (12 marks)

(a) Explain the general principle of **boosting**. How does it differ fundamentally from bagging? *(3 marks)*

(b) Describe how **AdaBoost** trains its sequence of predictors. If an AdaBoost ensemble is **overfitting**, what could you change? *(4 marks)*

(c) Describe how **gradient boosting** works (e.g. GBRT). What does the `learning_rate` hyperparameter control, and what is the relationship between the learning rate and the number of trees needed (shrinkage)? *(5 marks)*

Question 5. (12 marks)

(a) Explain how **stacking** differs from a simple voting classifier, and define the **blender** (meta-learner). *(4 marks)*

(b) When training a stacking ensemble, why are the blender's training features generated using **cross-validated (out-of-sample) predictions** from the base predictors rather than their predictions on the data they were trained on? *(4 marks)*

(c) Write a short scikit-learn snippet that builds a **soft-voting** classifier combining a `LogisticRegression`, a `RandomForestClassifier`, and an `SVC`. Note any change needed to the `SVC` for soft voting to work. *(4 marks)*